

Emotion Detection Learning Support Engine

Abstract

The Emotion Detection Learning Support Engine is an AI-powered application that detects a user's emotional state from text using a fine-tuned DistilBERT model and provides personalized guidance through the Gemini API. The system also stores prediction history and visualizes analytics using Streamlit and Plotly.

Objective

To build an intelligent learning support system that recognizes emotions from textual input and provides supportive recommendations, while maintaining prediction history and analytics.

Technology Description

- Frontend: Streamlit
- Language: Python
- NLP Model: DistilBERT (fine-tuned for emotion classification)
- AI Guidance: Google Gemini API
- Database: SQLite
- Data Processing: Pandas, NumPy
- Visualization: Plotly
- Model Libraries: PyTorch, Transformers

Detailed Execution of Project Phases

Phase 1 - Environment Setup

Configured Python virtual environment, installed required packages, and integrated Gemini API.

Phase 2 - Dataset Preparation

Prepared emotion dataset with train, validation and test splits and trained emotion classification models.

Phase 3 - Model Integration

Integrated the trained DistilBERT model for emotion prediction and keyword enhancement.

Phase 4 - AI Guidance

Generated personalized guidance using Gemini based on detected emotion and selected category.

Phase 5 - Database & Analytics

Stored prediction history and displayed dashboards showing emotion distribution and confidence.

Phase 6 - User Interface

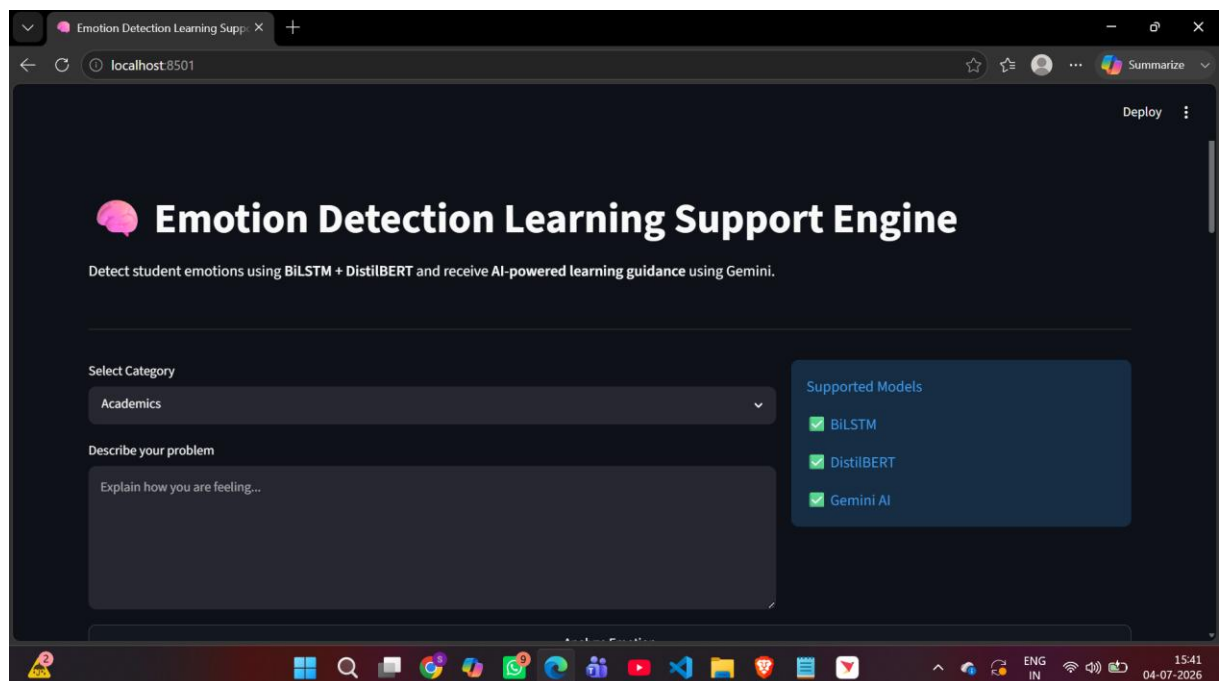
Developed a Streamlit web interface for emotion analysis, history, and analytics.

Project Explanation with Real-World Example

Example: A student selects 'Academics' and enters 'I am nervous about my final exams.' The system predicts 'Fear' with high confidence, stores the prediction, and generates supportive study guidance using Gemini. The user can later review the prediction history and analytics dashboard.

Screenshots

Home Page



Prediction History

Emotion Detection Learning Supp...

localhost:8501

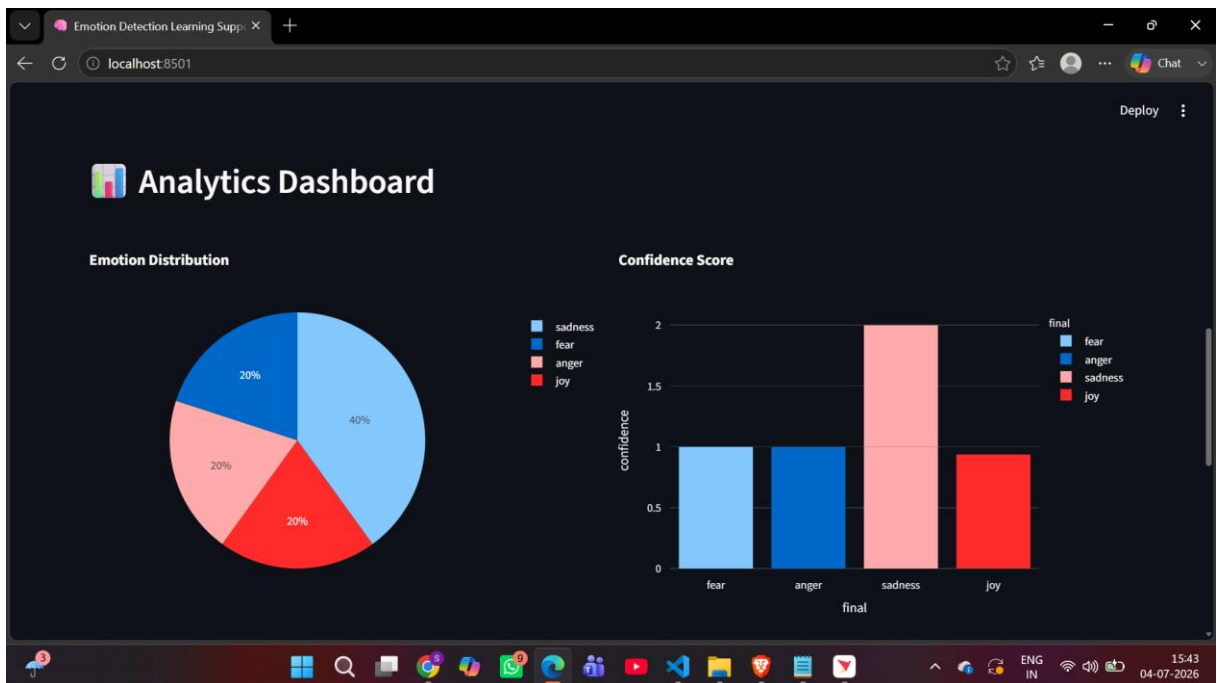
Analyze Emotion

Prediction History

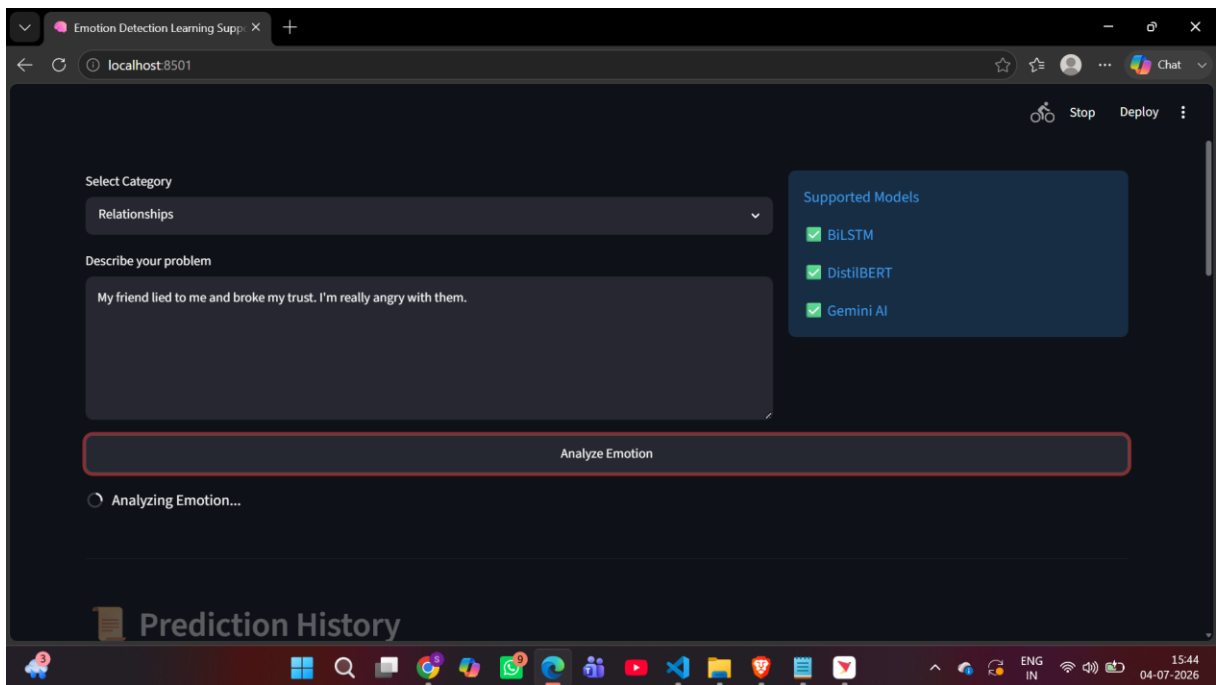
text	bert	final	confidence
I am nervous about my exam	fear	fear	0.9989
she is scolding on me	anger	anger	0.9981
My best friend stopped talking to me and I feel completely alone.	sadness	sadness	0.9995
My best friend stopped talking to me and I feel completely alone.	sadness	sadness	0.9995
My friend unexpectedly organized a birthday party for me. I never saw it coming.	joy	joy	0.9368

15:42 04-07-2026

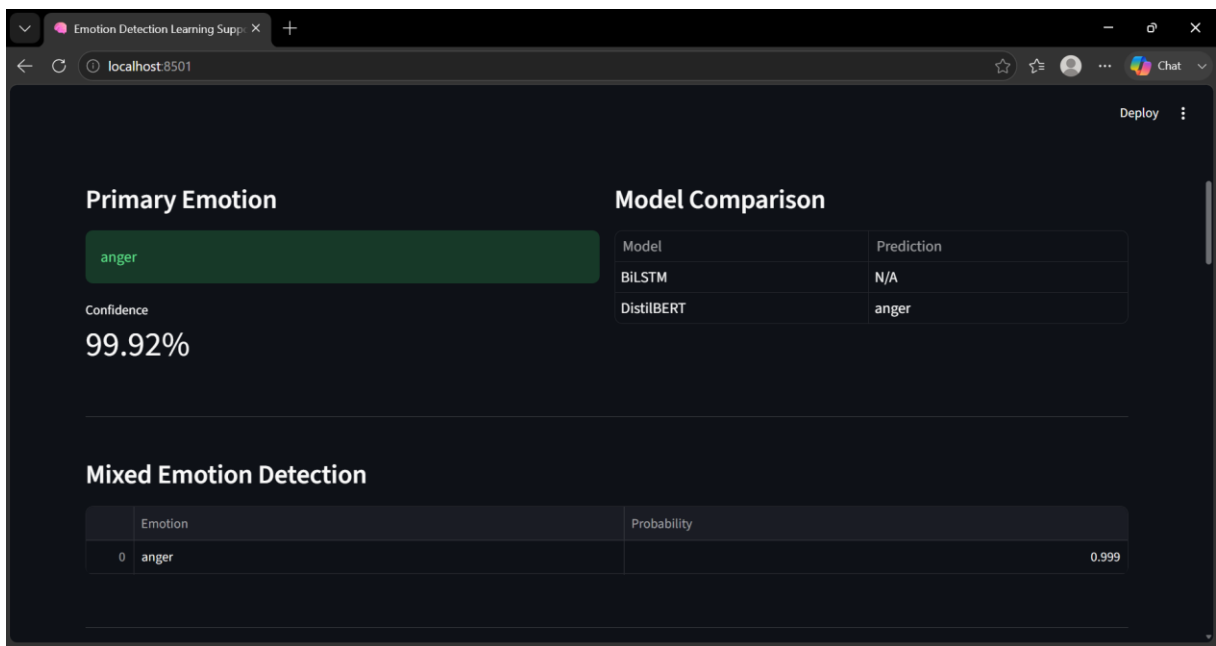
Analytics Dashboard



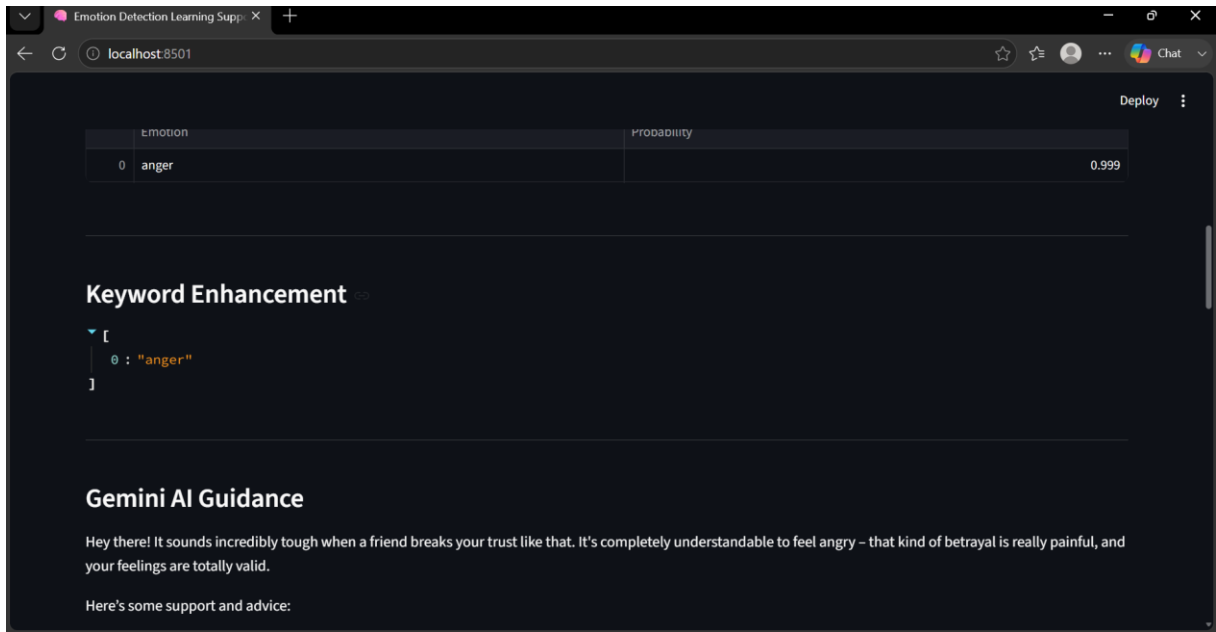
Input Screen



Prediction Result



Gemini AI Guidance



Conclusion

The Emotion Detection Learning Support Engine successfully combines modern NLP, AI guidance, analytics, and an interactive interface to provide emotion-aware learning support. The project demonstrates practical application of transformer-based emotion detection and generative AI for personalized assistance.